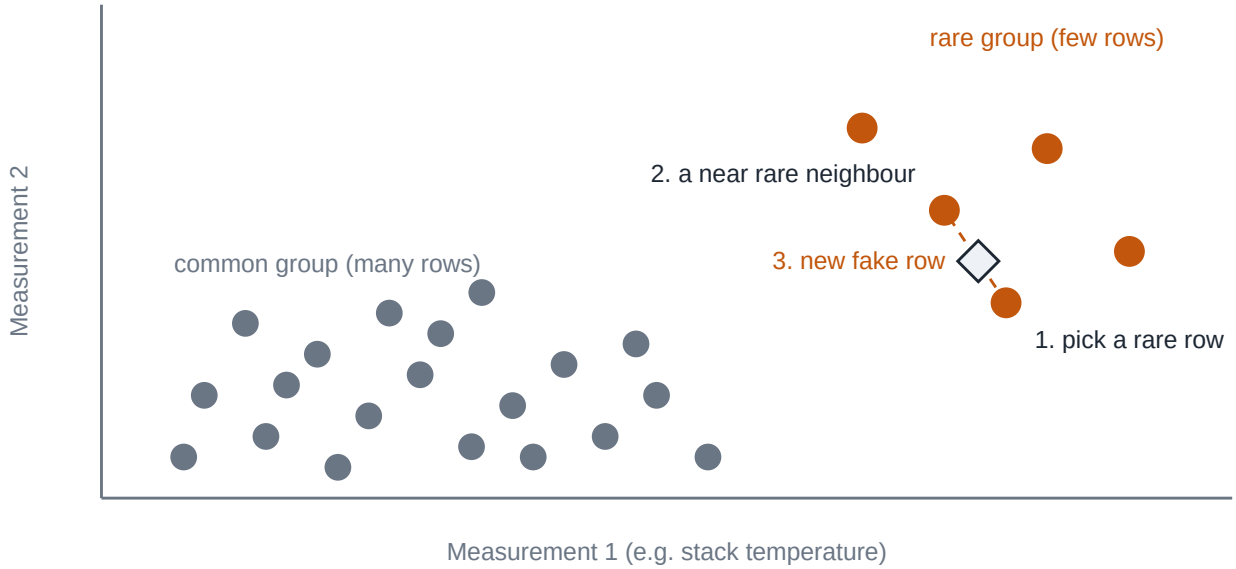
SOFC data · fake data with SMOTE

SMOTE Readiness Guide

> To use SMOTE (a method that invents extra rows for a rare group by mixing two real rows of that group), you need five things: a clear job for the model, one labelled row per example, a test set locked away first, enough real examples of the rare group, and checks that throw out impossible fake rows.

Why it matters: fake rows only help if they teach the model something true. Without these ingredients they either teach the model things the plant never does, or they leak into the test and make the scores look better than they are.

How SMOTE makes one fake row



Every fake row lies on a straight line between two real rare rows. SMOTE never goes outside what it has already seen.

The short list

What you need	In plain words	Your project today
A job for the model	What question should the model answer? (e.g. which stage is the plant in, will it trip soon)	**Not yet**
Labels on every example	Each example says which group it belongs to	**Have it** stage labels from the approved stage rules
One row per example	Curves cut into time windows and summed up as numbers	**Not yet**
A locked test set	Some whole runs kept aside before any fake rows are made	**Have it** 5 run-based test folds (F1 to F5)
Enough rare examples	At least 6 real examples of the rare group, from different runs	**Partly** Part load 10 runs, full Shut-down only 5
A check for impossible rows	Plant limits that every fake row must respect	**Partly** limits known from the stage notebooks

The next three pages go through these one by one.

- **What you need, step by step**: the full recipe and how many real examples you have
- **Why curves are tricky**: why SMOTE mixes up plant runs, and the other methods
- **Checking the fakes**: the three tests a fake row must pass, and your three decisions

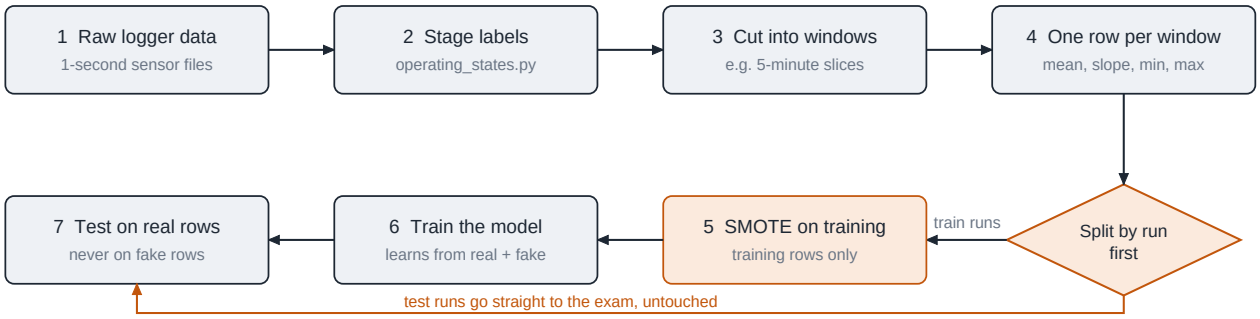
IF YOU REMEMBER ONE THING

SMOTE balances a training table by mixing rows you already have. It cannot create plant behaviour that was never recorded.

What SMOTE Needs

The order matters most: label the data, cut it into rows, set the test runs aside, and only then make fake rows, using the training part alone.

The recipe, in order



Seven steps. The orange diamond is the gate that matters: split by run before any fake rows are made.

Each ingredient, and where you stand

Need	Why	Your project today
1. A job for the model	SMOTE balances groups. You must know which groups, and that depends on the question.	**Not yet** to decide (see page 4)
2. Labels	SMOTE only mixes rows with the same label.	**Have it** 6 stages from the approved stage rules; trip events found in part load
3. One row per example	SMOTE works on a table. A 1-second curve has to be cut into windows (e.g. 5 minutes) and summed up: average, slope, lowest, highest.	**Not yet** not built yet
4. Clean numbers	SMOTE measures distance between rows. Gaps break it, and big-unit columns (temperature in °C) drown small ones (voltage).	**Partly** logger dropout minutes still to blank or fill; scaling still to add
5. A locked test set	Rows from one run look alike. If one run sits in both training and test, the test is too easy.	**Have it** folds F1 to F5 are split by run
6. Enough rare examples	The usual SMOTE setting looks at 5 neighbours, so it needs at least 6 real rare examples.	**Partly** see the table below
7. Plant limits	Used to throw out fake rows the plant could never produce.	**Partly** list on page 4
8. Software	The standard tool is the Python package imbalanced-learn.	**Not yet** not installed on this PC yet
9. A yardstick	Train the same model with and without fake rows and compare on the real test runs.	**Not yet** set up once the job is chosen

How many real examples do you have?

Stage	Real examples	Enough for SMOTE?
Hot standby	about 20 holds	**Enough**
Thermal adjustment	18 episodes	**Enough**
Start-up	15 episodes (8 cold, 7 warm)	**Enough** but cold and warm differ, so treat them apart
Part load	10 runs, 8 of them end in a trip	**Borderline** just enough; only 2 calm endings
Shut-down, full cool-down	5	**Borderline** too few for the usual setting; use 3 neighbours or fewer
Rated power	0	**None** SMOTE cannot make it; it only mixes what exists

Counting tip: count runs, not rows. Ten part-load runs cut into 5-minute windows give many rows, but SMOTE mixing two windows of the same run mostly copies that run.

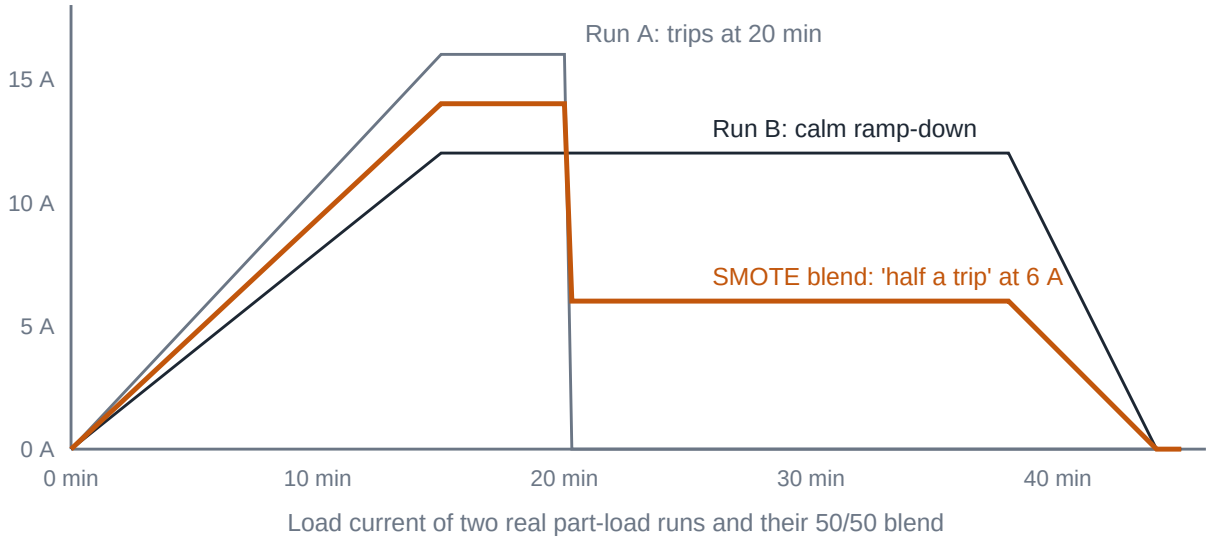
IF YOU REMEMBER ONE THING

Split the runs first. Make fake rows only inside the training part. Never test on fake rows.

Why Curves Are Tricky

SMOTE treats each row on its own, but a plant run is a story in time. Mixing two runs blends two stories into one that never happened.

What a blend looks like



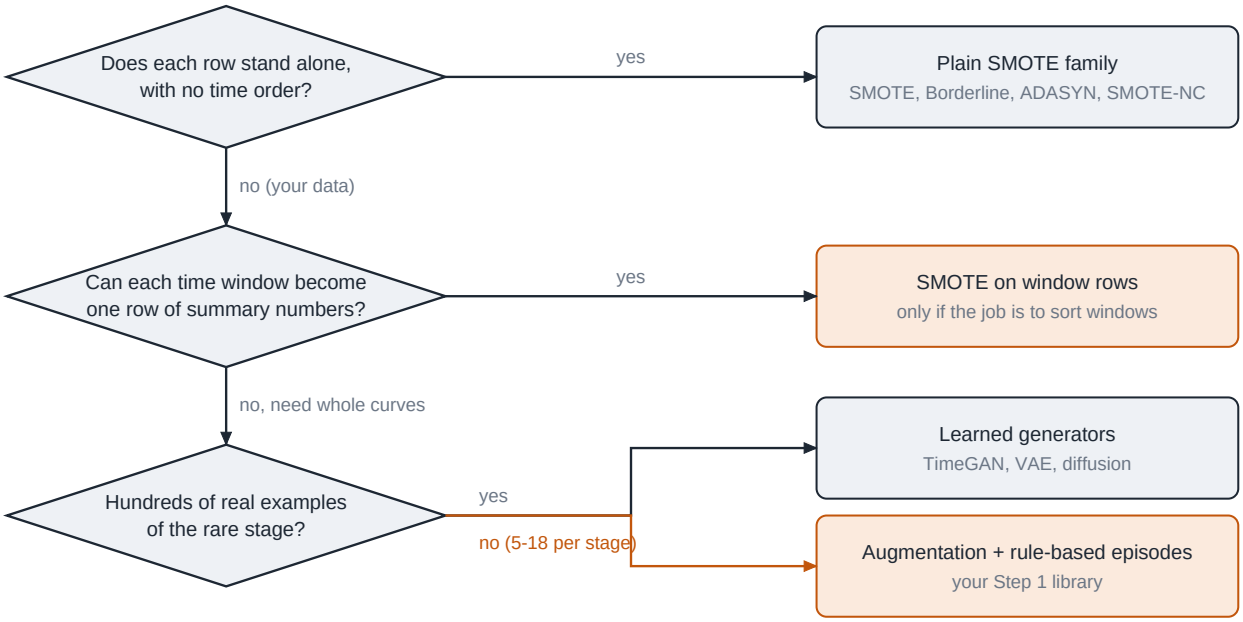
Two real part-load runs (grey and dark) and their blend (orange). The blend shows a 'half trip' that no plant can do.

The same happens with on/off signals. A heater that is on in one row and off in the other becomes 'half on' in the blend. And because SMOTE ignores time order, it can mix a minute from early heat-up with a minute from late heat-up.

The methods side by side

Method	What it does	Needs	Fit for your data
SMOTE	Mixes two rare rows	A table of numbers	**Limited** only on window summary rows
SMOTE-NC	Same, but copies on/off and text columns instead of averaging them	A mixed table	**Limited** use this one if on/off columns stay in
Borderline-SMOTE, ADASYN	Puts more fake rows near the border between groups	A table, low noise	**Limited** optional variant, try after plain SMOTE
Small changes to real curves	Adds a little noise, scales or stretches a real run in time	Real curves	**Good fit** simple, keeps the run's order
Learned generators (TimeGAN, VAE, diffusion)	A model learns to draw brand-new curves	Hundreds to thousands of runs	**Poor fit** far too few runs today
Rule-based episodes	Stitches real stage episodes in the allowed order and varies them within limits	Stage labels and rules	**Good fit** your current plan, Step 1 done

Which one to pick



Follow the questions from the top. Your data takes the orange paths.

IF YOU REMEMBER ONE THING

Use SMOTE on summary rows to balance a classifier. Use small changes and the rule-based plan to make whole new runs.

Checking the Fakes

A fake row is kept only if it passes three tests: it obeys the plant's limits, it looks like the real data, and it makes the model better on real test runs.

The three gates

Each fake row goes left to right. Failing any orange gate means drop it or rethink the method.

Gate 1: plant limits

Rule	Limit	Where it comes from
Heat-up speed	heater outlet ramp about 60 K/h; stack about 50 °C/h	Start-up notebook
Lowest cell voltage	not below 0.72 V while under load	operator limit seen in part load
Load current	up to 16.2 A, the highest seen	Part-load notebook (higher only if you decide on design values)
Exhaust temperature	at 200 °C or more, a trip must follow	trip analysis
On/off signals	exactly on or off, never in between	plant logic
Stage order	only the 18 allowed arrows of the stage map	stage map

Gate 2: does it look real?

Compare the spread of each sensor in real and fake rows. Then train a small 'spot the fake' model. If it can only guess right about half the time, the fakes look real.

Gate 3: does it help?

Train the same model twice on the same folds F1 to F5: once on real rows only, once with fake rows added. Score both on the real test runs. Keep the fake rows only if the score goes up. A moderate top-up (for example the rare group raised to half the size of the common one) is often safer than full balance.

Three decisions only you can make

Decision	Options	Suggested start
1. What job should the model do?	find the current stage; warn before a trip; estimate a sensor value	warn before a trip (trips are the rare, costly event)
2. How much to top up the rare group?	a little, half, or full balance	half, then test

Decision	Options	Suggested start
3. SMOTE, the rule-based plan, or both?	SMOTE for a classifier on window rows; rules for whole new runs	both, each for its own job

IF YOU REMEMBER ONE THING

Fake rows earn their place only by passing the plant limits, looking real, and improving results on real test runs.